



VISUAL ANALYTICS: UNCERTAINTY AND BIAS

AB MOSCA (THEY/THEM)



NOTES

- Mid-semester project (mp) due Thursday
- hw04 released Thursday, but you have plenty of time to work on it before/after break

TODAY'S PLAN

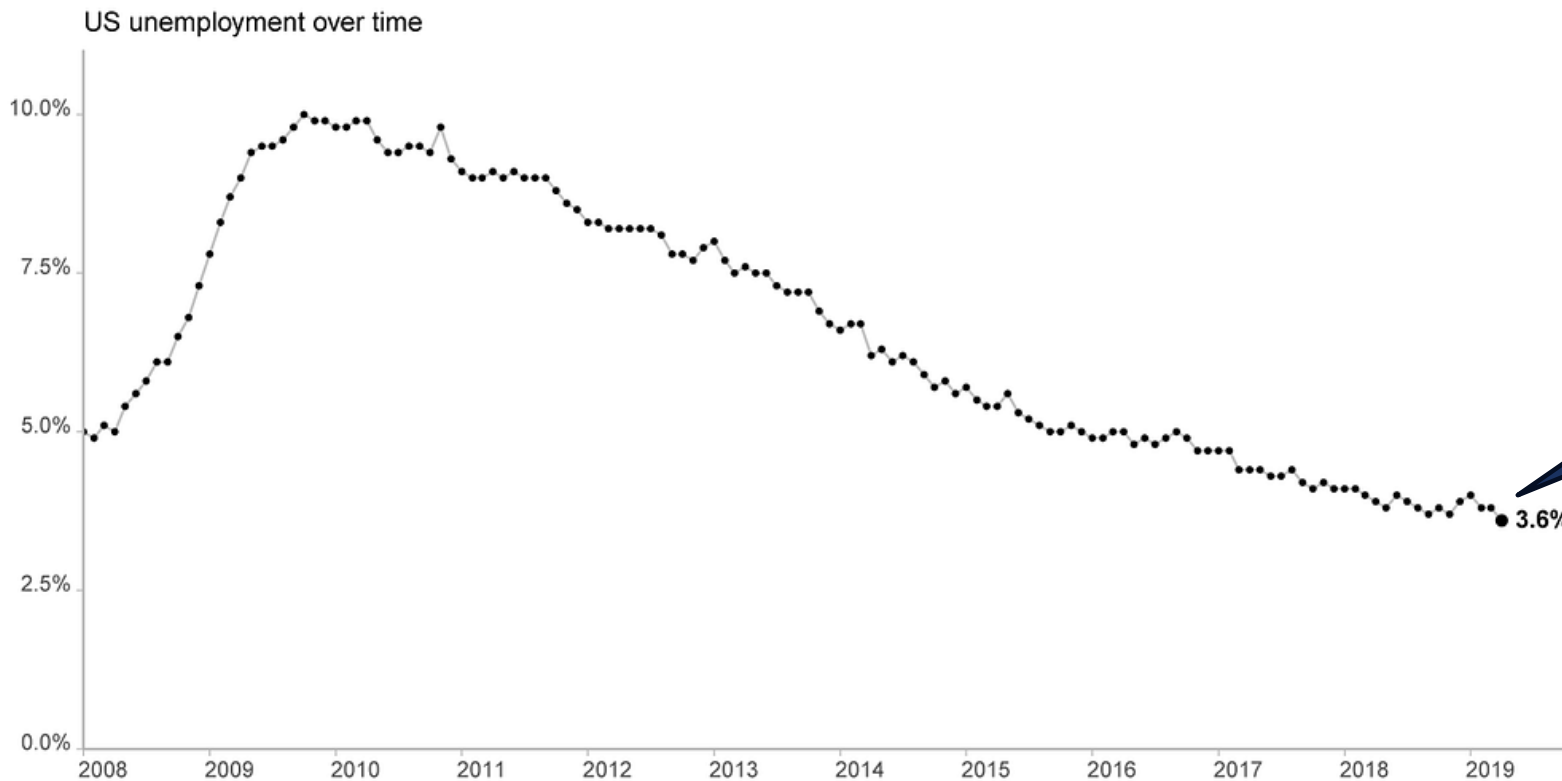
- Uncertainty
 - Definition
 - Visualizing
- Bias
 - Definition
 - Visualizing

DEFINITION: UNCERTAINTY

- Three types of uncertainty:
 - Aleatoric: irreducible randomness inherent in a process
 - Epistemic: uncertainty from a lack of knowledge that could theoretically be reduced given more information
 - Ontological: uncertainty about how accurately the modeling describes reality, which can only be described subjectively
- In visualization, when we refer to uncertainty, we typically mean quantified uncertainty that can be visualized

DEFINITION: UNCERTAINTY

(Visualization based on data from the Bureau of Labor Statistics)



How is this value calculated?

DEFINITION: UNCERTAINTY

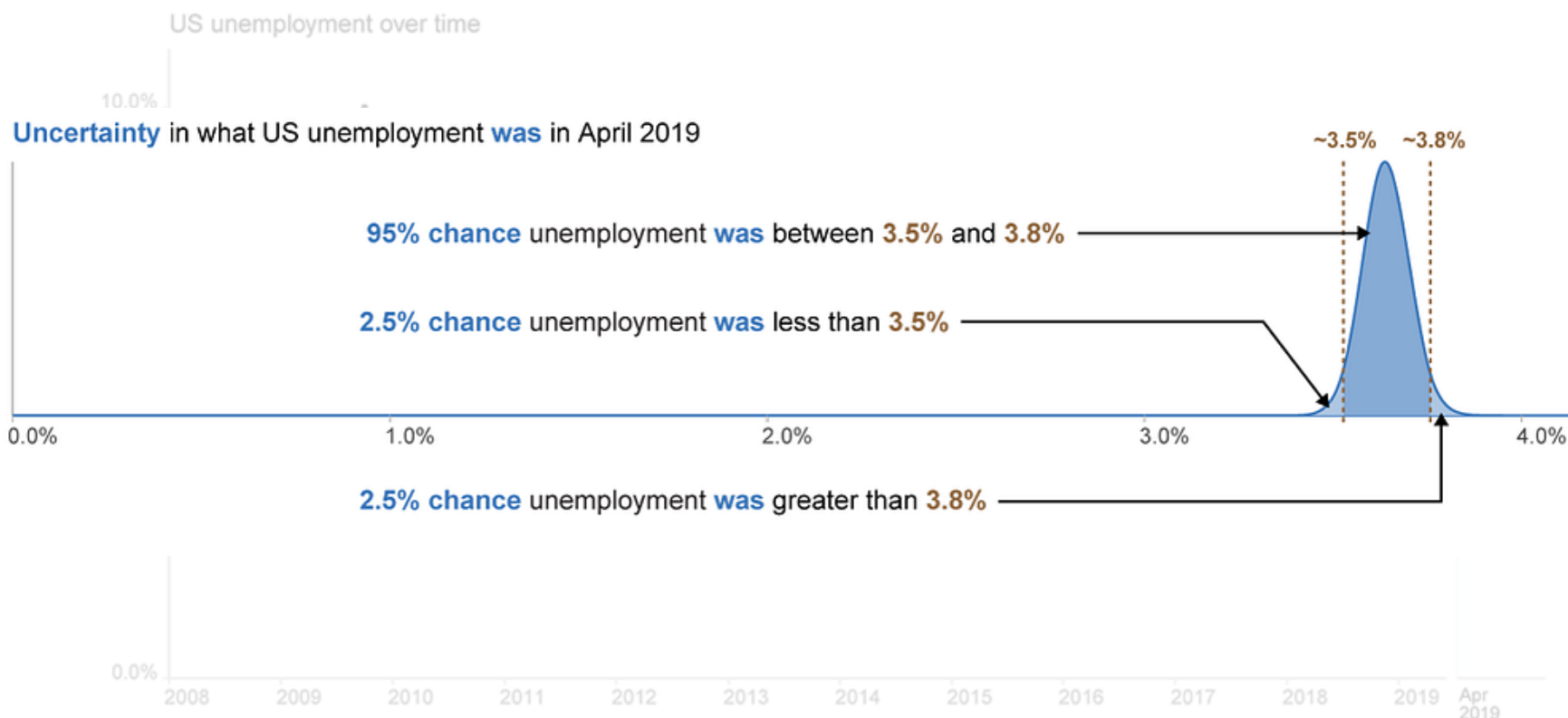
(Visualization based on data from the Bureau of Labor Statistics)



- A statistical model is used to estimate a parameter
- The model quantifies the uncertainty around the estimate by considering all values it could be (and how likely each is)
- Hence, a parameter's uncertainty can be characterized by a probability distribution

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VISUALIZING UNCERTAINTY

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Why is it important to communicate uncertainty?

VISUALIZING UNCERTAINTY

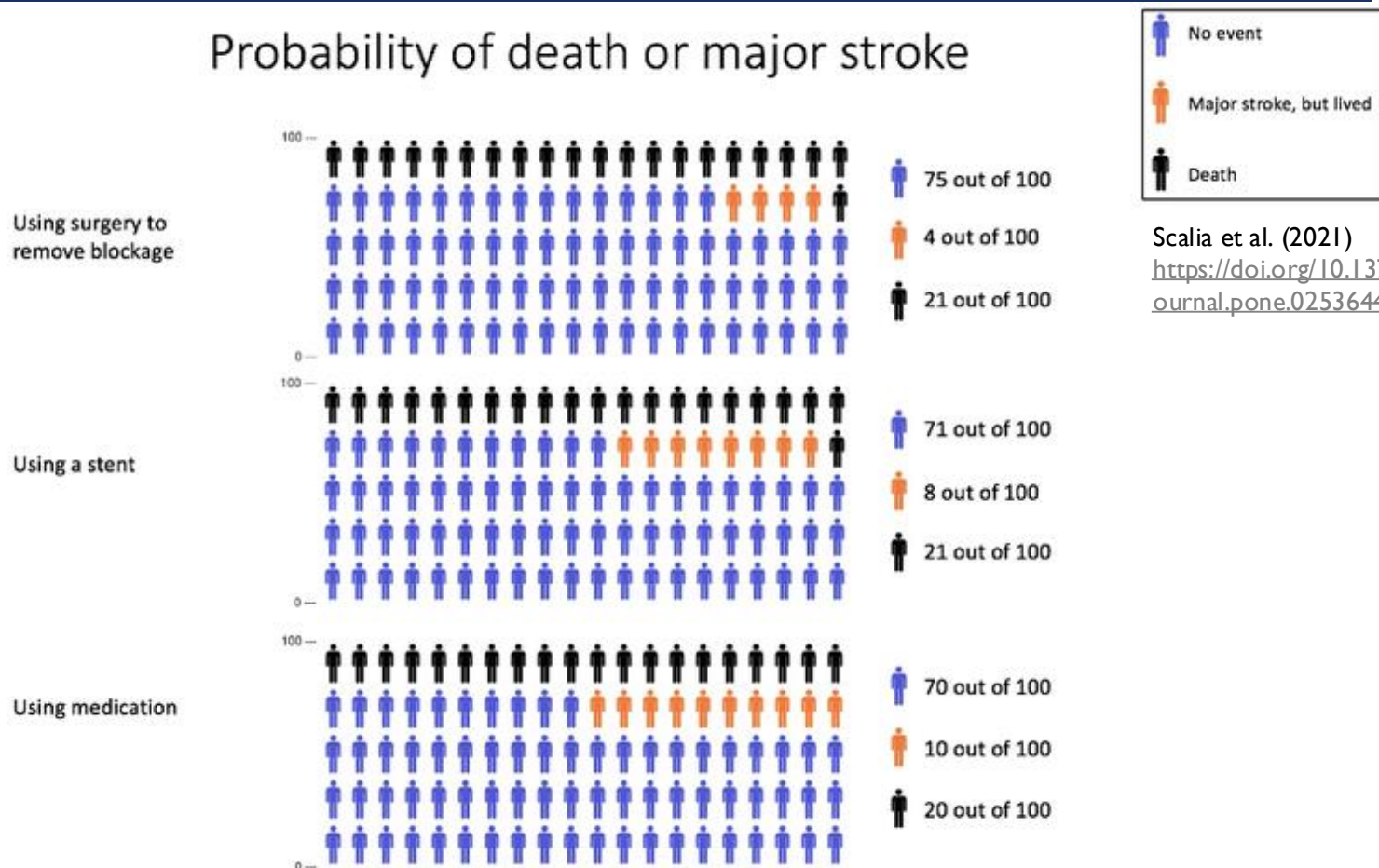
- There are two broad categories of uncertainty visualization techniques:
 1. Graphical annotations that show properties of a distribution
 - Ex. error bars, animating between possible values
 2. Mapping probability or confidence to a visual encoding channel
 - Ex. Make a mark more transparent if uncertainty is high

VISUALIZING UNCERTAINTY

Icon Array

- Offload cognition by allowing a viewer's visual system to compare the denominator and the numerator in a frequency probability format
- Visual comparisons of this nature are easier and faster than numerical calculations
- Can reduce denominator neglect by allowing people to compare relative ratios visually

Probability of death or major stroke

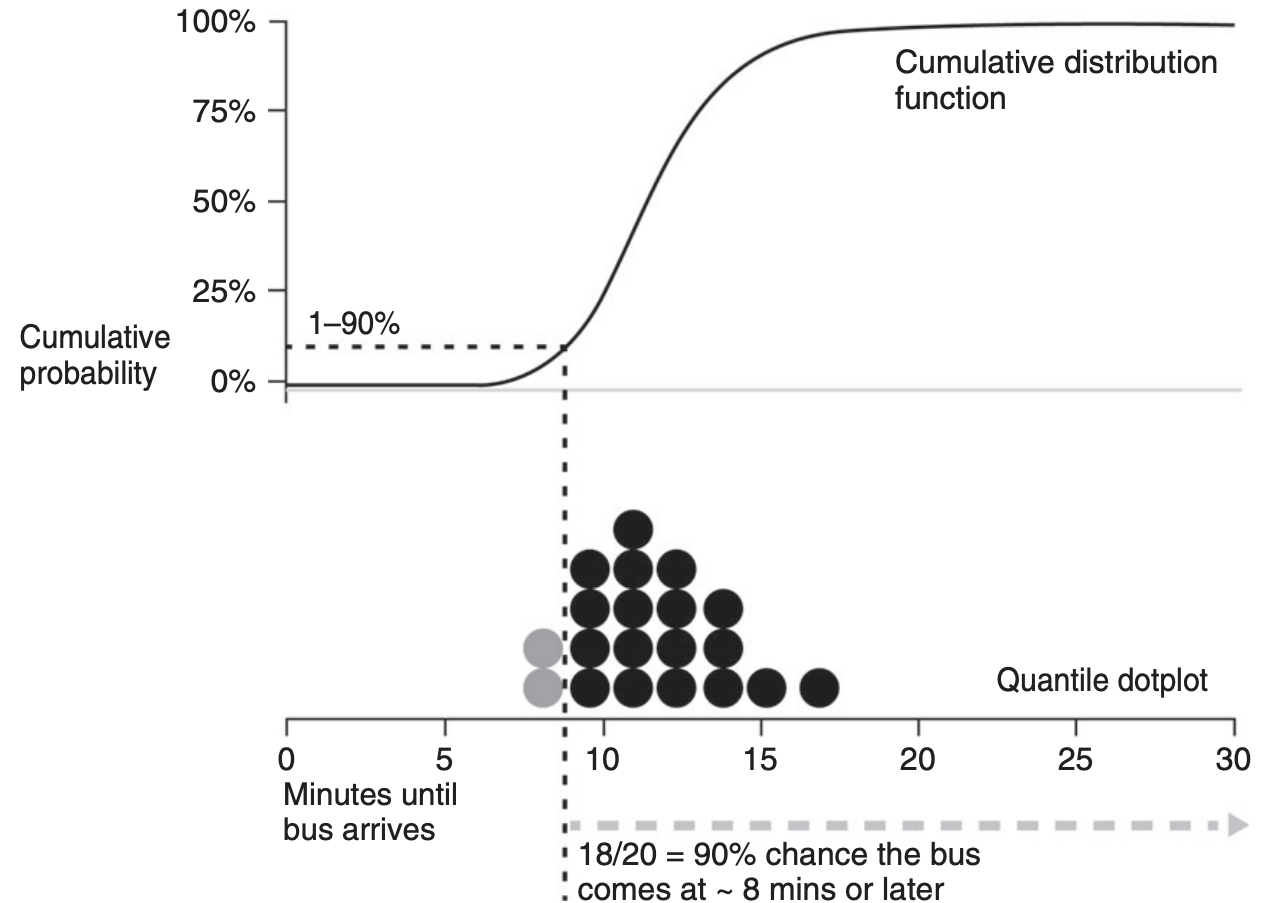


Scalia et al. (2021)
<https://doi.org/10.1371/journal.pone.0253644>

VISUALIZING UNCERTAINTY

Quantile dotplot

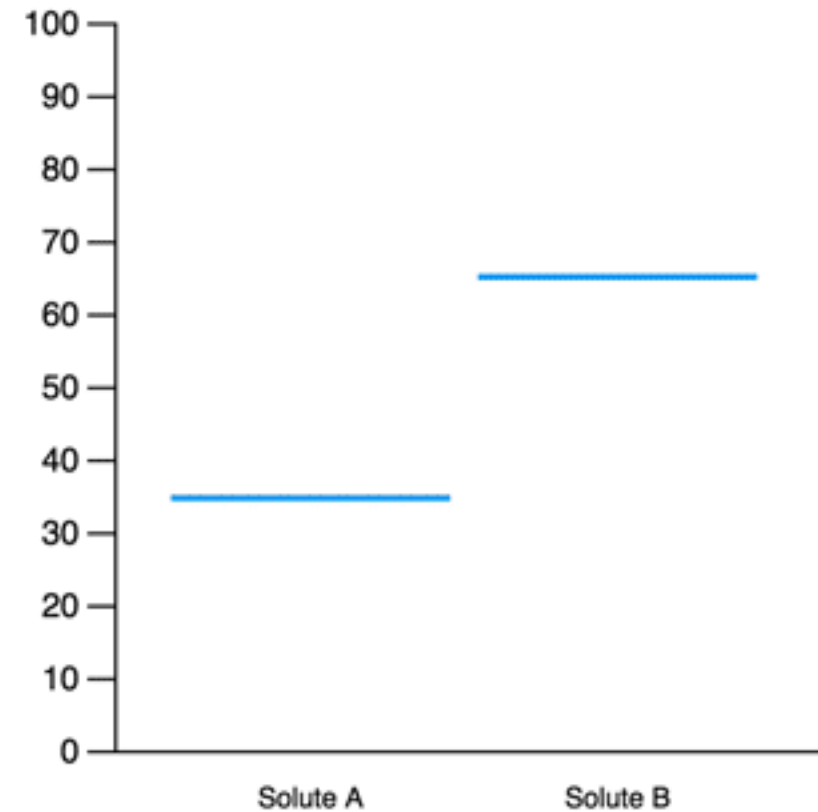
- Displays uncertainty for a continuous variable
- Represents a distribution where dots are sampled proportional to the quantiles of the distribution



VISUALIZING UNCERTAINTY

Hypothetical outcome plots (HOPs)

- Use random draws from a distribution and animates the draws over time
- Force viewers to:
 - account for uncertainty in their understanding of the data
 - recognize that less probable outcomes do fall within the distribution
- Can be applied to most distribution data and visual encoding techniques relatively easily (as long as animation is not already used)

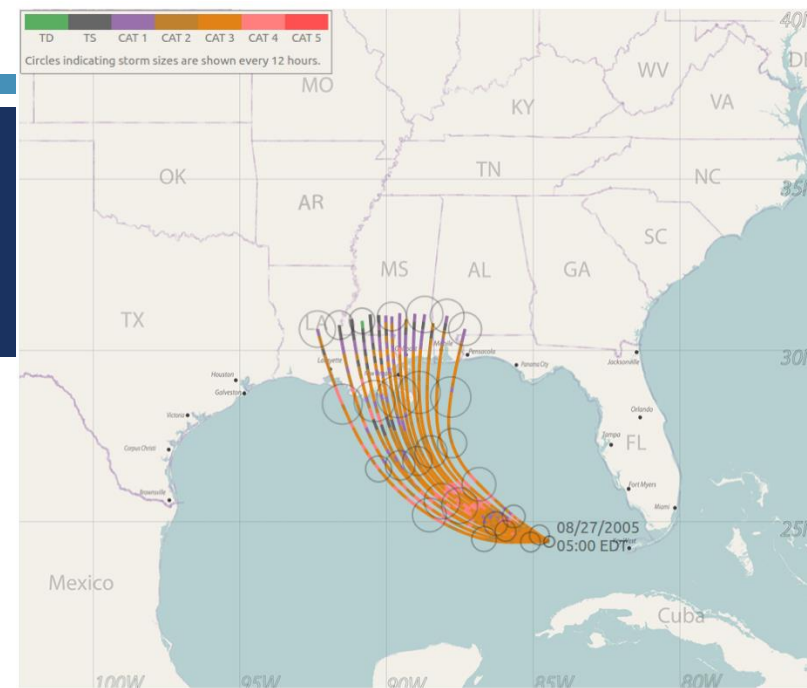


<https://medium.com/hci-design-at-uw/hypothetical-outcomes-plots-experiencing-the-uncertain-b9ea60d7c740>

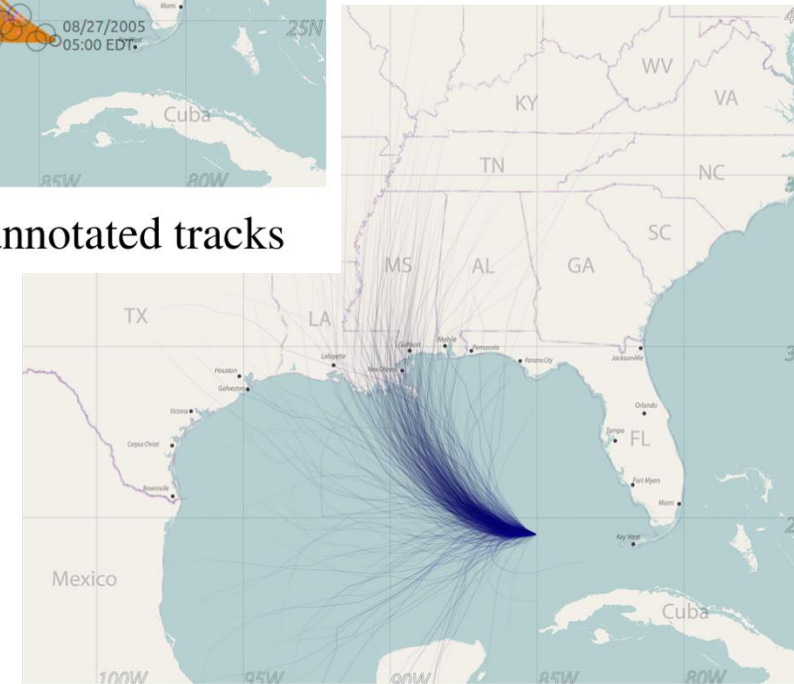
VISUALIZING UNCERTAINTY

Ensemble display

- Typically used to illustrate potential storm paths
- Traditionally generated by making perturbations to a model's parameters and plotting the resulting runs on a static display
- Studies show people overreact when one model appears to be hitting a point of interest vs. near a point of interest



(c) selectively sampled annotated tracks



(b) implicit: ensemble of tracks

VISUALIZING UNCERTAINTY

Error bars, Box plot

- Visual boundaries force people to think differently about the data than a continuous encoding, (visual boundaries=cognitive categories) which can be good or bad, depending on the nature of their decision



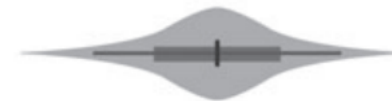
Error bars



Box plot

Violin plot, Gradient plot

- Violin plots (mapping probability to area or width at a given y position) and gradient plots (mapping probability to opacity) lead to more intuitive assessments of value likelihood and “ability to surprise”



Violin plot

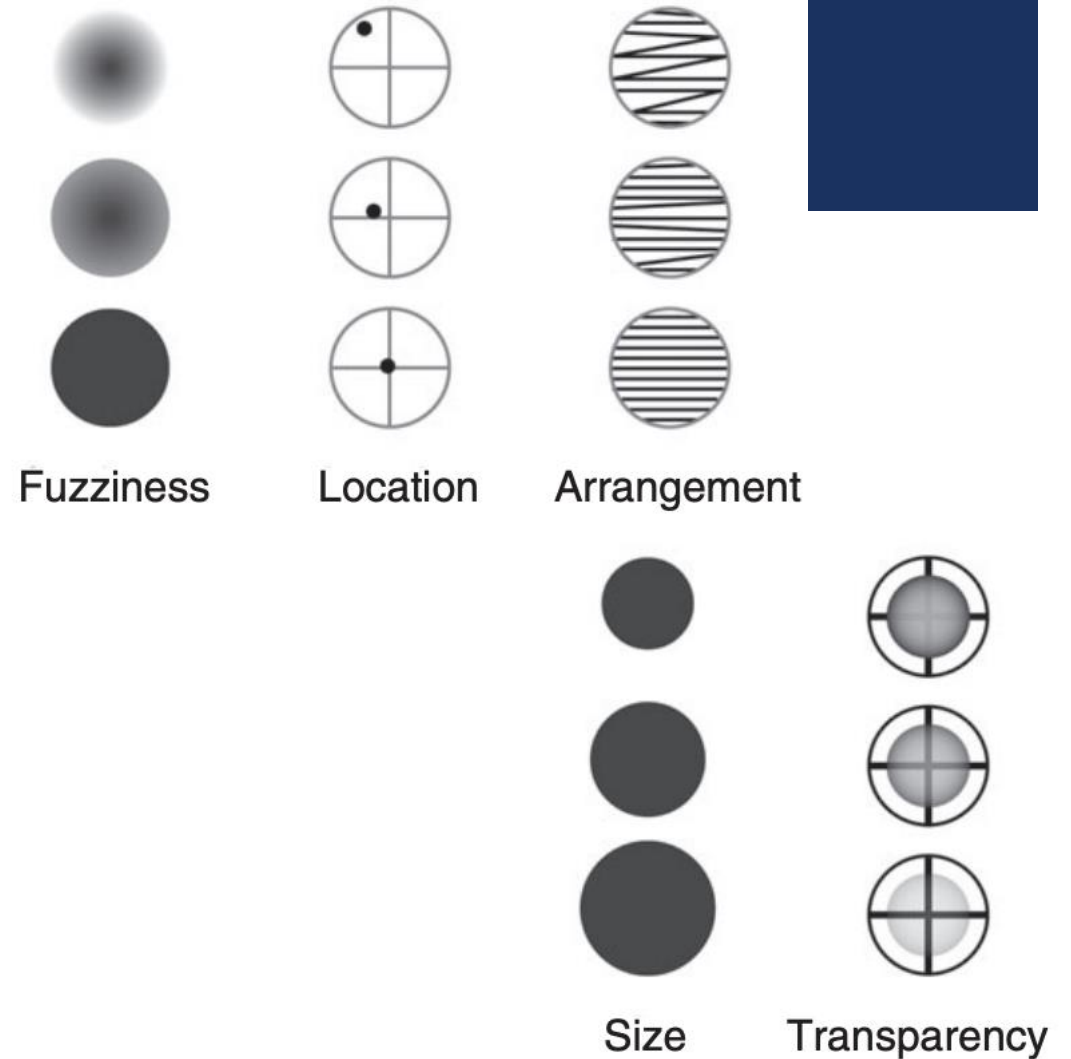


Gradient plot

VISUALIZING UNCERTAINTY

Fuzziness, transparency, location, etc.

- Encoding techniques that utilize visual metaphors for uncertainty, such as graying out using color saturation, out of focus using blur, fogginess using transparency, adding noise using texture, or sketchiness, are more intuitive ways to communicate uncertainty
- Some of the visualizations map probability to visual properties and are designed to inhibit the viewer from resolving the value of a datum when the uncertainty is too high



VISUALIZING UNCERTAINTY

Value-suppressing color pallets

- Attempt to make perception difficult in proportion to uncertainty
- Blends hues denoting value with gray proportional to how uncertain they are. As a result, the most uncertain values all appear as the same shade of gray

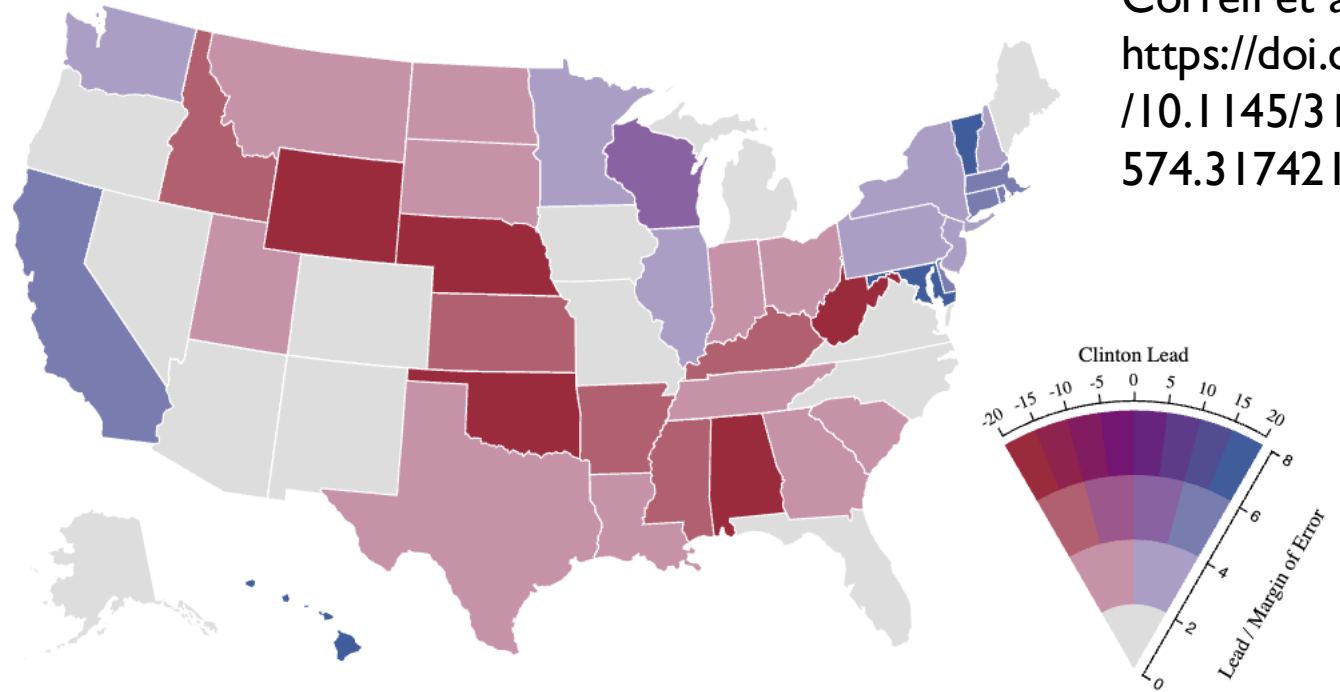


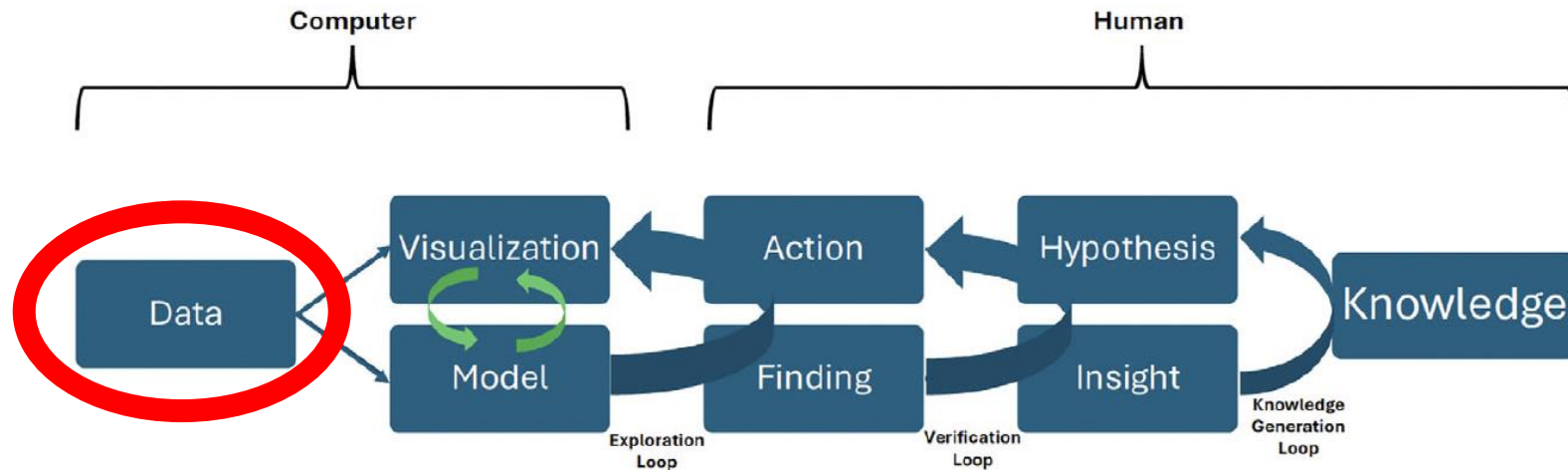
Figure 5: Polling data prior to the 2016 U.S. Presidential Election, encoded with a VSUP.

YOUR TURN

Are these methods used in the “real world”?

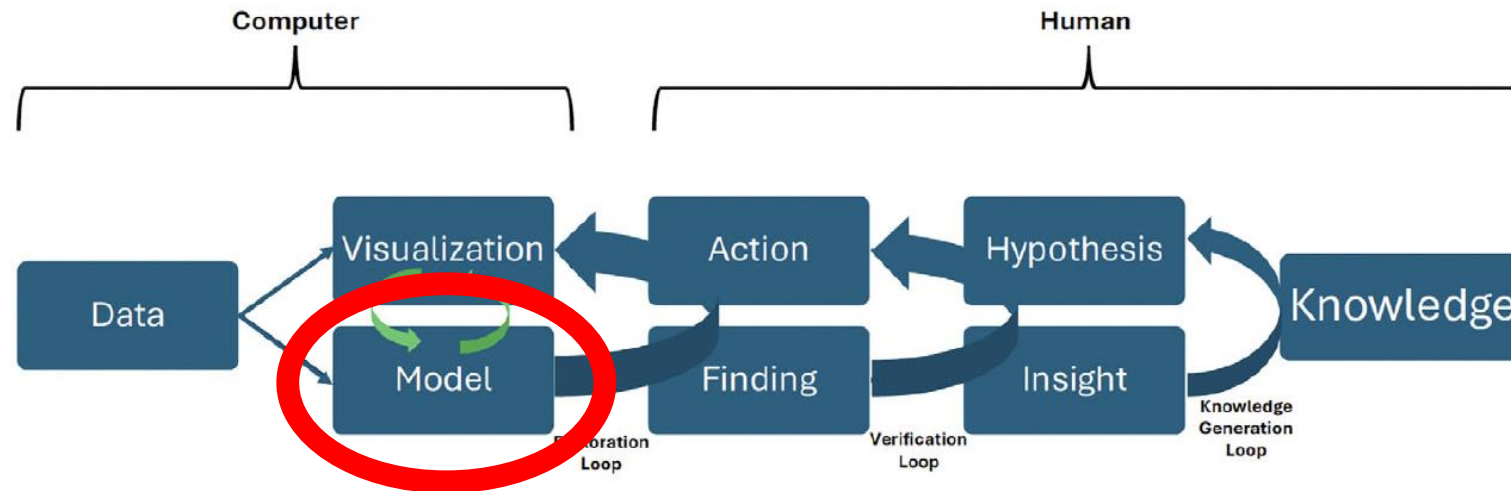
- Find a visualization on your favorite news source
- Does the visualization show uncertainty?
 - If so, which method does it use?
- How could you re-design the visualization to show uncertainty (or to show it better)?

BIAS IN VISUAL ANALYTICS



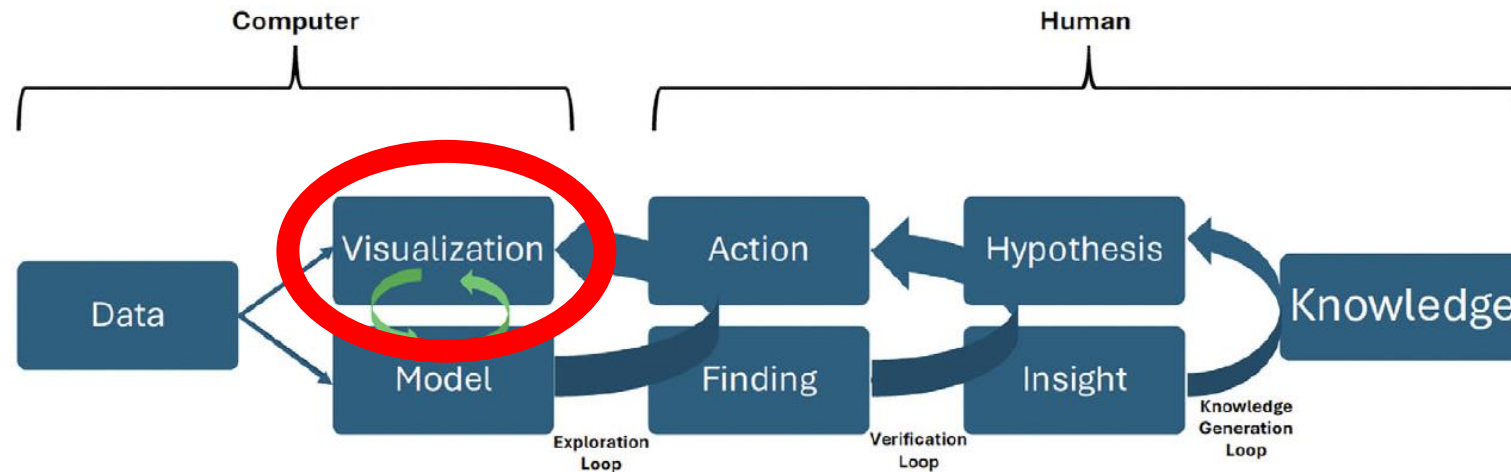
- Measurement precision (ex. effect of gerrymandering on voting records)
- Completeness (ex. aggregated or missing values)
- Historical bias (ex. hiring decisions, arrest records)
- Sampling bias (ex. convenience sampling on college students)

BIAS IN VISUAL ANALYTICS



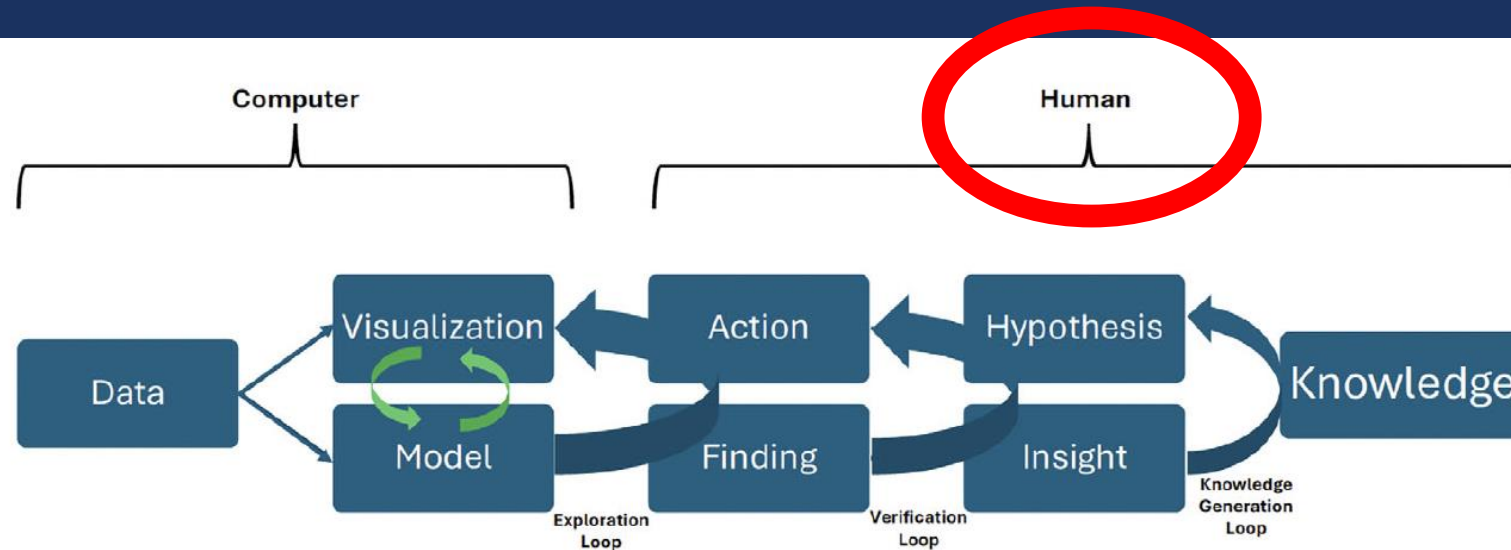
- Simplifications and assumptions about training data
- Aggregation bias (one model used for heterogeneous groups)
- Evaluation bias (in appropriate training benchmarks)

BIAS IN VISUAL ANALYTICS



- Perceptual errors
 - Aspect scale influencing rate of change
 - Comparing lengths vs areas
 - Poor estimation of proportions
- Assuming causation given a scatter or bar plot

BIAS IN VISUAL ANALYTICS



Three perspectives relevant to visual analytics:

1. bias as a cognitive processing error
2. bias as a filter for information
3. bias as a preconception

BIAS IN VISUAL ANALYTICS

Bias as a cognitive processing error

- Two cognitive processes:
 - Intuition - largely subconscious and prone to making errors (bias)
 - Reason - intentional deliberation, responsible for recognizing and correcting errors
- Cognitive processing errors result from shortcuts in cognition, broadly referred to as heuristics. Bias then is described as the method or mechanism by which the error occurs
- Note: the process of heuristic decision-making does not always lead to errors; it usually facilitates fast decision-making

BIAS IN VISUAL ANALYTICS

Bias as a cognitive processing error

- **Confirmation bias** - people tend to accept confirmatory evidence of a pre-existing hypothesis and dismiss contrary information
- **Availability bias** - people tend to rely more heavily on information that is easily remembered (e.g., most recent)
- **Attraction effect** - tendency for a decision to be influenced by an inferior alternative
- **Anchoring bias** - tendency to be heavily reliant on an initial value or anchor

Describe a scenario in which one of these biases could affect a visual analytic analysis

BIAS IN VISUAL ANALYTICS

Bias as a filter for information

- Information overload, refers to a point beyond people's cognitive and perceptual limits where performance and decision-making suffer
- Under overload conditions, people selectively allocate attention and other mental resources to the tasks or information of highest priority. One's filter or bias thus determines what information is gathered and how sensory information is distinguished and interpreted
- Ex. Participants were shown video footage of people wearing either white or black shirts passing a basketball and asked to count how many times white-shirt basketball players on a team passed the ball. Most participants count the appropriate number of passes but **about half fail to perceive a glaringly misfit player walk across the court.**

Describe a scenario in which this bias could affect a visual analytic analysis

BIAS IN VISUAL ANALYTICS

Bias as a **preconception**

- Analysts' analysis are unconsciously shaped by experiences and internal influences
- Unconscious bias may arise from
 - cultural beliefs and traditions
 - general self-confidence or self-esteem
 - comfort and familiarity with a machines analytics and interface functions
 - amount of risk aversion vs. seeking

Describe a scenario in which one of unconscious biase
could affect a visual analytic analysis

BIAS IN VISUAL ANALYTICS

Asynchronous interventions

- Education (ex. about unconscious biases)

Big Limitation:

Education of biases requires that analysts *remember* information about biases and recall what strategies to apply to mitigate them. In reality, analysts may forget to engage the information in real-time decision-making scenarios

BIAS IN VISUAL ANALYTICS

Real-Time interventions

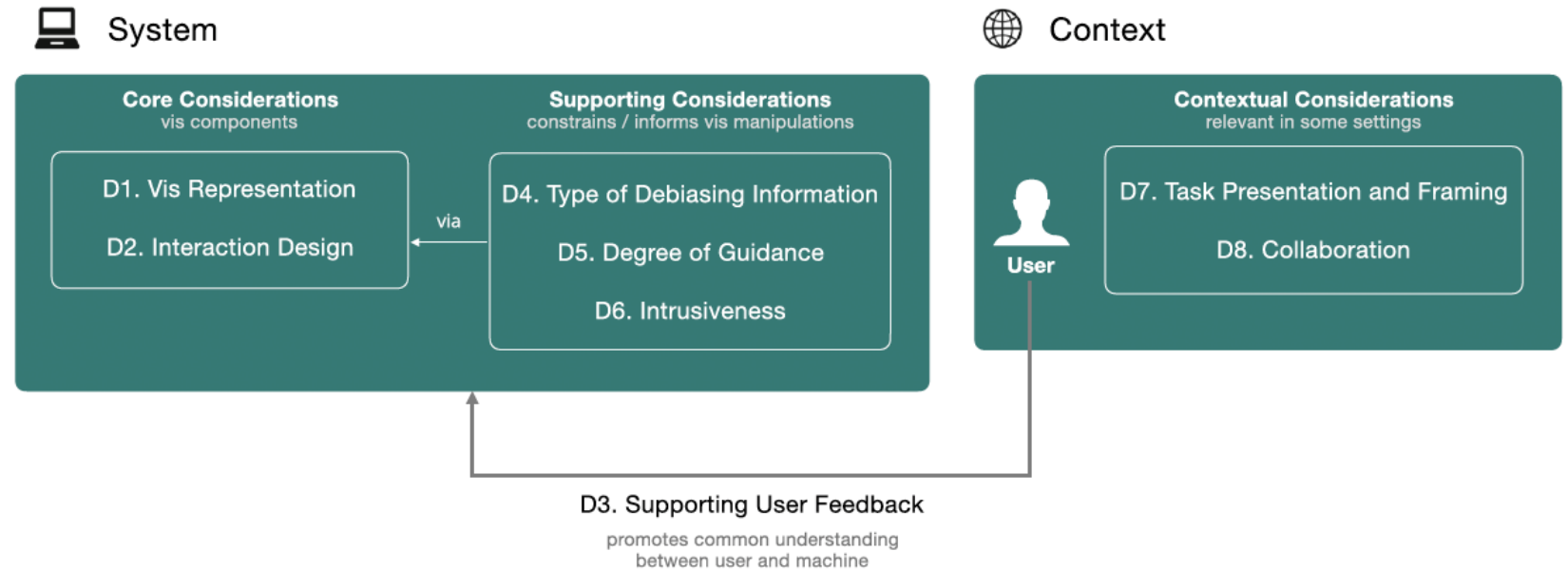


Fig. 7.1 The design space is comprised of 8 dimensions, described in Sect. 7.3.2. D1 (VISUAL REPRESENTATION) and D2 (INTERACTION DESIGN) are the two *core* components of a visualization [43] that can be manipulated to mitigate biased decision-making processes. How these components are manipulated is informed and constrained by *supporting* considerations, including D4 (TYPE OF DEBIASING INFORMATION), D5 (DEGREE OF GUIDANCE) and D6 (INTRUSIVENESS). Some *contextual* considerations may only be relevant in specific settings, including D7 (TASK PRESENTATION AND FRAMING) and D8 (COLLABORATION). Finally, D3 (SUPPORTING USER FEEDBACK) connects the user and contextual setting to the system by promoting a common understanding between user and machine

BIAS IN VISUAL ANALYTICS

Real-Time interventions

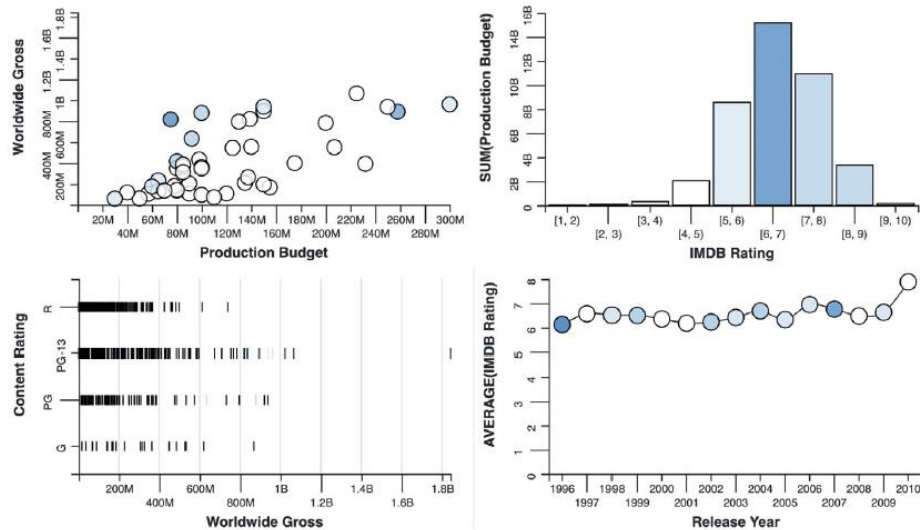


Fig. 7.5 Lumos captures mouseover interactions on visualization elements (e.g., points, bars) and leaves traces in-situ by coloring (fill) them in shades of blue in proportion to the relative frequency of interactions

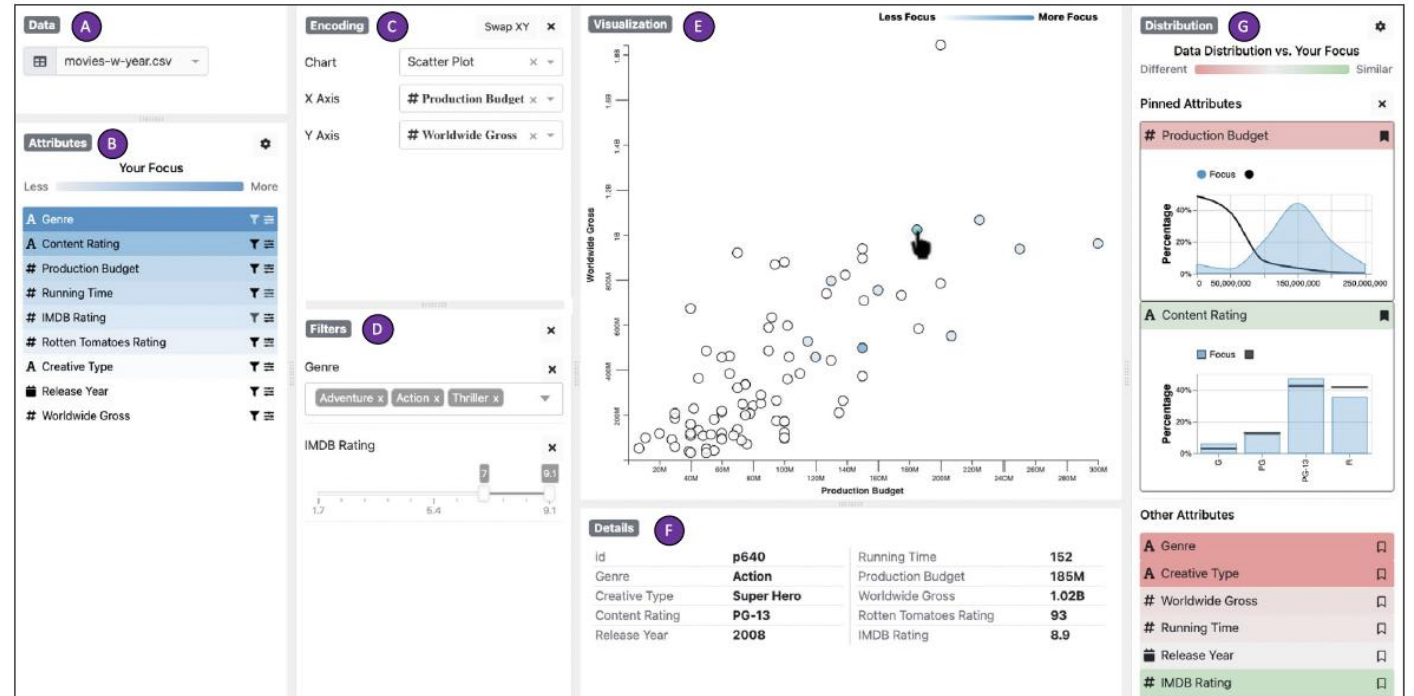


Fig.7.3 The Lumos user interface includes traditional visual data analysis functions ((A) **Data Panel**, (B) **Attributes Panel**, (C) **Encoding Panel**, (D) **Filter Panel**), and shows analytic behavior as in-situ and ex-situ interaction traces in the (B) **Attributes Panel**, (E) **Visualization Canvas**, (F) **Details View**, and (G) **Distribution Panel** as the relation between a target distribution (e.g., the underlying data) and the user's analytic behavior